Click Captcha Recognition Pipeline

Input Image
[H, W, 3]

YOLOv8n
Object Detector

Detection Boxes [N, 6] (x1, y1, x2, y2, conf, cls)

Hint / Candidate Splitter
3 strategies: Edge - Geometric - UI Container [see (a)]

Hint Crops
[N_hint, H, W, 3]

Candidate Crops
[N_cand, H, W, 3]

ArcFace Embedding Extractor xN
ResNet18 Backbone + L2 Normalize -> 256-dim [see (b)]

Hint Embeddings [N_hint, 256] | Candidate Embeddings [N_cand, 256] (L2-norm)

Cosine Similarity S = H · C^T

Similarity Matrix [N_hint, N_cand]

Greedy Global Matching

Output: Click Positions [(cx1, cy1), (cx2, cy2), ...]

(a) Hint / Candidate Splitter

Edge Strip
Scan 15px border edges
Group tightly-packed boxes
Score by alignment quality

Geometric
Find nearly-arranged group
Common Y/X coordinate
Consistent size & spacing

UI Container
Threshold (gray > 195)
Morphological close + open
Find light panel contour

Priority Decision Tree

UI ∩ Geometric same set? -> use it | UI finds 3-5 boxes? -> use UI | Edge finds 3-5 boxes? -> use Edge
Else -> Geometric | Fallback -> largest group (2-6 boxes)

hint_indices (sorted)

candidate_indices

(b) ArcFace Embedding Extraction (per crop)

xN
Crop Image
[H', W', 3]

Letterbox
112×112

ResNet18
Backbone

L2
Normalize

256-dim
Embedding

Legend

Tensor | Module | Operation | Output

Main flow | Detail flow | xN badge